October 2026

ChemRxiv Editorial Team

Re: Submission of "Beyond Candidate Generation: A Validation-First Protocol for AI-Assisted Transition-State Discovery"

Dear Editors,

I am pleased to submit the manuscript above for consideration as a ChemRxiv preprint. The manuscript is a methods and perspective contribution on how to validate transition-state (TS) candidates produced by AI models. It is not a claim of a new predictive model; it is a protocol that separates candidate generation from physical validation and makes the validation layer reproducible.

The central argument is that a generated geometry is a hypothesis, not a transition state. A TS requires one imaginary frequency, the intended atom pair, the correct connectivity, and a consistent electronic state. The manuscript presents a four-stage protocol—chemical preflight, barrier-region scan, constrained reference plus Hessian, and validation—and implements it as a portable, open-source skill package for local ORCA/xtb and Gaussian/Slurm execution. It illustrates the protocol with internal case studies from Ni/Fe catalysis, including a per-point scan strategy that completed 16/16 points after a collective scan failed three times, and a generative-model test on an 84-atom broken-symmetry organometallic system where the candidates were out of distribution.

I believe the work is well suited to ChemRxiv because it is a community-facing methods contribution: the protocol and the implementation are meant to be used and criticized before journal submission. The preprint will also establish a citable reference for the open-source package. The manuscript is not under consideration elsewhere, and I am the sole author.

The author declares no competing financial interest. The code repository is in preparation; the package is available from the corresponding author and will be linked in the final version.

Suggested categories: Computational Chemistry (primary); Catalysis (secondary).

Sincerely,

Hao-Fan Li
State Key Laboratory of Structural Chemistry
Fujian Institute of Research on the Structure of Matter
Chinese Academy of Sciences, Fuzhou 350002, China
University of Chinese Academy of Sciences, Beijing 100049, China
[corresponding email]